

Animesh Saha

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EDUCATION

University at Buffalo, The State University of New York Buffalo, New York
Bachelor of Science, Computer Science — Specialization in Artificial Intelligence Expected December 2025
– GPA: 3.85, Summa Cum Laude | Tau Beta Pi Inductee (Spring 2025) | Dean's List (Spring 2025, Spring 2024, Fall 2023)

SKILLS

Languages: Python, Java, OCaml, SQL, JavaScript, TypeScript, HTML/CSS
Frameworks/Tools: React.js, Next.js, FastAPI, LangChain, PyTorch, Docker, Qdrant, FAISS, sentence-transformers, Google Gemini API, Firebase, MySQL, Pandas, NumPy
Concepts: RAG & Vector Search, LLM Evaluation, REST API Design, Data Structures & Algorithms, Computer Vision, Deep Learning, OAuth 2.0/Authentication, Agile Development

EXPERIENCE

Software Engineer (Capstone) August 2024 – December 2024
Capstone Project
– Engineered a mood-based music recommendation system with a full-stack REST API and React frontend, improving recommendation accuracy by 60%.
– Integrated OAuth 2.0 authentication and MySQL to automate Spotify playlist generation; system served 4,500 users and was architected to support 2,500+ concurrent sessions.

Undergraduate Teaching Assistant — CSE 305: Programming Languages August 2025 – December 2025
Dept of CSE, University at Buffalo
– Supported 250 students across CSE 305 by debugging complex programming projects, teaching functional programming and language-theory concepts, and grading assignments; held weekly office hours to resolve technical questions.

Academic Tutor August 2024 – December 2025
Tutoring and Academic Support Services, University at Buffalo
– Tutored 60+ students one-on-one and staffed drop-in tutoring for 600+ students in Data Structures, Algorithms, and Calculus 2, visualizing algorithms (BFS, DFS, Dijkstra, A*) through whiteboard walkthroughs and live coding sessions.

Student Assistant August 2023 – December 2025
Special Programs Management, School of Engineering and Applied Science, University at Buffalo
– Screened 50+ applicants and conducted interviews to place qualified candidates into open positions; separately led a team of 5 to organize 3 Dean's Advisory Council meetings per semester.

PROJECTS

RAG System over arXiv Research Papers | *Python, Docker, Qdrant, LangChain, FastAPI* July 2026
– Built an end-to-end Retrieval-Augmented Generation (RAG) ingestion pipeline over 200+ arXiv research papers, chunking abstracts and full-text PDFs into 2,344 retrievable chunks, embedded with sentence-transformers and indexed in a Dockerized Qdrant vector database.
– Engineered a hand-wired retrieval-to-generation loop integrating the Gemini API, achieving a 77.8% retrieval hit-rate@5 on an 18-question hand-labeled evaluation set, and built a dual LLM-as-judge/human evaluation harness that uncovered a self-grading ceiling-effect bias in the LLM judge.
– Refactored the pipeline with LangChain (LCEL) while holding retrieval logic identical via a custom retriever wrapper, producing a quantified framework-vs-hand-wired tradeoff analysis, and diagnosed a payload-schema bug in LangChain's default vector-store integration before it caused silent citation data loss.
– Built a FastAPI backend exposing the pipeline through a Pydantic-validated REST endpoint and a Next.js/React/TypeScript chat frontend rendering grounded answers with clickable citations to source papers.

Ice Breaker — Real-Time Social Platform | *Next.js, React, Firebase* November 2025
– Built a real-time, anonymous group-chat web app used live by 1,000+ attendees at a large in-person social gathering, requiring no login or personal data storage.
– Integrated Gemini LLM to summarize conversations and answer contextual questions within group chats.
– Optimized Next.js/React architecture, reducing latency from 500ms to under 200ms in high-traffic conditions.

Draw It Right — Winner, UB Hacking 2024 | *Python, PyTorch, OpenCV, Mediapipe* November 2024
– Designed a finger-tracking, gesture-based virtual drawing tool using OpenCV/Mediapipe, and integrated a PyTorch neural network to classify user-drawn objects in real time with 92.5% accuracy.
– Deployed the app on Streamlit for an interactive, user-friendly interface, winning UB Hacking 2024.